

COVID infection rates and unmet mental-health need - core analysis

Dataset background, licences, research questions and the full analysis plan are in [Project_Proposal.qmd](#). This file holds the runnable core analysis that answers the two research questions.

We want to know whether places affected harder by COVID also reported more unmet need for mental-health support. We measures one row per **US state** per **survey window**. The script below builds that table in three steps

- the mental-health outcome
- a comparable COVID case rate
- join them and then answers the two research questions.

```
library(readr)
library(dplyr)
library(tidyr)
library(ggplot2)

# The outcome is a single switch so the whole analysis can be re-run for a
# different indicator (must match an Indicator value in the Household Pulse CSV).
OUTCOME_INDICATOR <- "Needed Counseling or Therapy But Did Not Get It, Last 4 Weeks"
```

1. Mental-health outcome: unmet need by state × survey window

The Household Pulse file stores every indicator/subgroup/state/period combination in one long table. We keep only the rows we need, the chosen outcome, split *By State*, with a real value, and rename the survey-window dates to short names we will reuse everywhere. We build the same thing once at state level (for RQ1) and once at national level (for RQ2).

```

read_mh <- function(group, state_filter = NULL) {
  d <- read_csv("../Data/Mental_Health_Care_in_the_Last_4_Weeks.csv",
                show_col_types = FALSE) |>
    filter(Group == group, Indicator == OUTCOME_INDICATOR, !is.na(Value))
  if (!is.null(state_filter)) d <- filter(d, State == state_filter)
  d |>
    transmute(
      state      = State,
      win_start  = as.Date(`Time Period Start Date`, "%m/%d/%Y"),
      win_end    = as.Date(`Time Period End Date`,    "%m/%d/%Y"),
      unmet_need = Value                                     # % of adults reporting the outcome
    )
}

mh      <- read_mh("By State")                                # one row per state × window
mh_nat <- read_mh("National Estimate", "United States") # one row per window

cat("state x window rows:", nrow(mh),
    "| states:", n_distinct(mh$state),
    "| windows:", n_distinct(mh$win_start), "\n")

```

state x window rows: 1676 | states: 51 | windows: 33

2. COVID case rate per state, comparable across states

What's in the two new datasets

Both new sources are small, official and tidy, so a short look is enough. We use only a few columns from each (the COVID file's `deaths` and `fips` go unused).

File	Columns we use	Meaning
covid_us_states_nyt.csv	date, state, cases	day · US state · cumulative confirmed cases
us_state_population_2020.csv	state, population	US state · 2020 resident population

```

pop      <- read_csv("../Data/us_state_population_2020.csv", show_col_types = FALSE)
covid_raw <- read_csv("../Data/covid_us_states_nyt.csv",      show_col_types = FALSE)

```

```
glimpse(covid_raw)
```

```
Rows: 61,942
Columns: 5
$ date    <date> 2020-01-21, 2020-01-22, 2020-01-23, 2020-01-24, 2020-01-24, 20~
$ state    <chr> "Washington", "Washington", "Washington", "Illinois", "Washingt~
$ fips     <chr> "53", "53", "53", "17", "53", "06", "17", "53", "04", "06", "17~
$ cases    <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, ~
$ deaths   <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, ~
```

```
glimpse(pop)
```

```
Rows: 51
Columns: 2
$ state      <chr> "Alabama", "Alaska", "Arizona", "Arkansas", "California", "~
$ population <dbl> 5024279, 733391, 7151502, 3011524, 39538223, 5773714, 36059~
```

```
# Missing values in the columns we actually use (expect 0 in both):
covid_raw |> summarise(across(c(date, state, cases), ~ sum(is.na(.))))
```

```
# A tibble: 1 x 3
  date state cases
<int> <int> <int>
1     0     0     0
```

```
pop      |> summarise(across(c(state, population), ~ sum(is.na(.))))
```

```
# A tibble: 1 x 2
  state population
<int>         <int>
1     0           0
```

Why per 100k. Raw case counts would just rank states by size (California will always look worst). To compare a small state with a large one we convert to **new cases per 100,000 residents**, using the population table above (US Census Bureau, 2020 Decennial Census, 50 states + DC).

The NYT file reports *cumulative* cases. The quantity we actually want is *new* cases on each day, which is just today's total minus yesterday's.

```
covid_daily <- covid_raw |>
  arrange(state, date) |>
  mutate(new_cases = pmax(cases - lag(cases), 0), .by = state) |> # cumulative -> daily
  inner_join(pop, by = "state") |>
  mutate(new_per100k = new_cases / population * 1e5)
```

3. Join COVID intensity into each survey window

Each survey window is a date range (about two weeks). For every state we add up its daily `new_per100k` that falls inside that window. The COVID change a state experienced while that survey was running. A **range join** (`between(date, win_start, win_end)`) matches each COVID day to the window it belongs to, so no manual looping is needed.

```
covid_win <- covid_daily |>
  inner_join(distinct(mh, state, win_start, win_end),
    by = join_by(state, between(date, win_start, win_end))) |>
  summarise(covid_rate = sum(new_per100k), .by = c(state, win_start, win_end))

joined <- inner_join(mh, covid_win, by = c("state", "win_start", "win_end"))

cat("joined rows:", nrow(joined), "\n")
```

joined rows: 1676

```
summary(joined$covid_rate)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
6.687	129.981	272.419	451.207	571.719	5089.035

```
summary(joined$unmet_need)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
3.30	9.20	10.60	10.75	12.20	20.80

4. RQ1: Does COVID intensity relate to unmet mental-health need?

With one comparable pair of numbers per state-window, we fit a straight line: does unmet need go up as the COVID case rate goes up? The slope answers the question; its 95% confidence interval and p-value tell us how sure we are.

```
m1 <- lm(unmet_need ~ covid_rate, data = joined)
summary(m1)
```

Call:

```
lm(formula = unmet_need ~ covid_rate, data = joined)
```

Residuals:

Min	1Q	Median	3Q	Max
-7.2449	-1.5453	-0.1423	1.4414	10.2448

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.051e+01	7.327e-02	143.433	< 2e-16 ***
covid_rate	5.434e-04	1.057e-04	5.142	3.03e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.277 on 1674 degrees of freedom

Multiple R-squared: 0.01555, Adjusted R-squared: 0.01496

F-statistic: 26.44 on 1 and 1674 DF, p-value: 3.034e-07

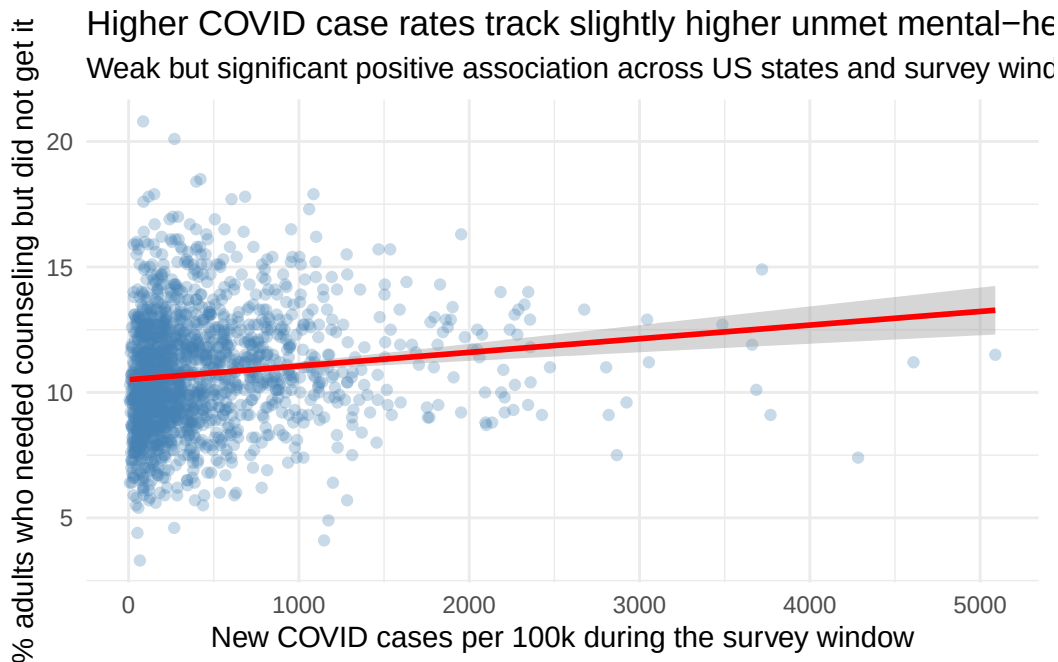
```
# Population claim -> report the coefficient with its 95% CI, not just the p-value.
cbind(estimate = coef(m1), confint(m1))
```

	estimate	2.5 %	97.5 %
(Intercept)	10.508850002	1.036515e+01	1.065255e+01
covid_rate	0.000543447	3.361607e-04	7.507332e-04

```
ggplot(joined, aes(x = covid_rate, y = unmet_need)) +
  geom_point(alpha = 0.3, colour = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, colour = "red") +
  labs(
    title = "Higher COVID case rates track slightly higher unmet mental-health need",
    subtitle = "Weak but significant positive association across US states and survey windows"
```

```
x = "New COVID cases per 100k during the survey window",
y = "% adults who needed counseling but did not get it"
) +
theme_minimal()
```

``geom_smooth()`` using formula = 'y ~ x'



5. RQ2: How did need and COVID co-move nationally over time?

RQ1 compares states; RQ2 instead follows the *whole country* through time. We build a national COVID case rate the same way (total new cases in each window ÷ US population) and plot it beside national unmet need on a shared time axis, so we can see whether the two rise and fall together across pandemic waves.

```
us_pop <- sum(pop$population)

covid_nat <- covid_daily |>
  inner_join(distinct(mh_nat, win_start, win_end),
    by = join_by(between(date, win_start, win_end))) |>
  summarise(covid_rate = sum(new_cases) / us_pop * 1e5, .by = c(win_start, win_end))
```

```

mh_nat |>
  left_join(covid_nat, by = c("win_start", "win_end")) |>
  select(win_start,
         `Unmet need (%)` = unmet_need,
         `COVID cases /100k` = covid_rate) |>
  pivot_longer(-win_start) |>
  ggplot(aes(win_start, value)) +
  geom_line(colour = "darkblue") +
  geom_point(size = 1) +
  facet_wrap(~name, ncol = 1, scales = "free_y") +
  labs(title = "National unmet mental-health need and COVID case rate over time",
       x = "Survey window start", y = NULL) +
  theme_minimal()

```

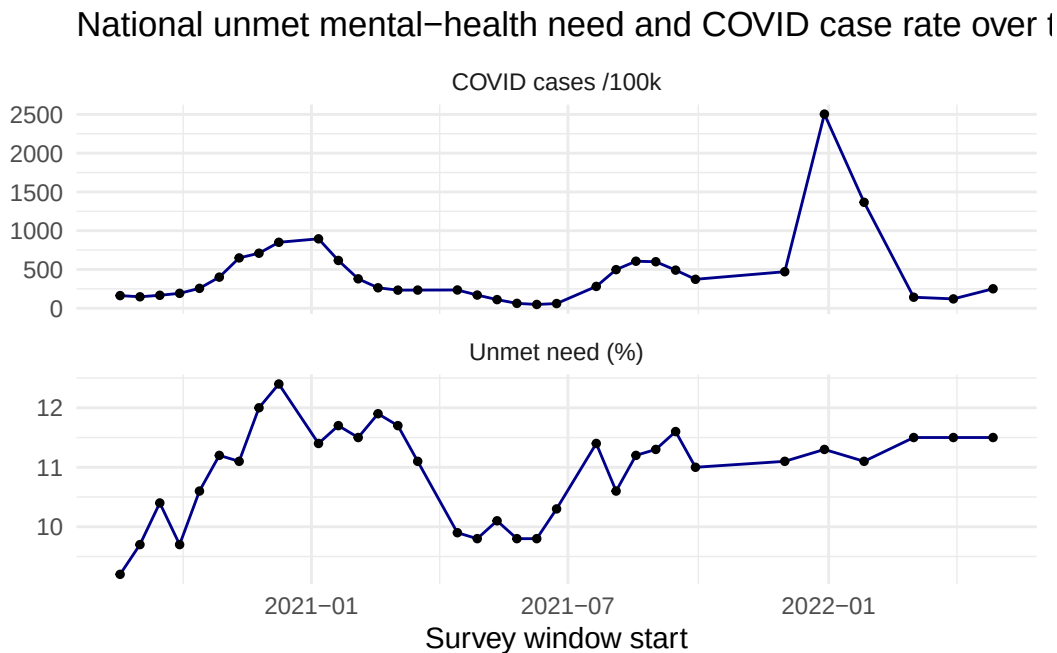


Figure 1: Two-panel time series of national unmet mental-health need (top) and the national COVID-19 case rate (bottom), by survey window, Aug 2020 to Apr 2022. Plotting them on a shared time axis shows whether the two series rise and fall together across pandemic waves.

Notes / next steps (for the remaining weeks)

- RQ1 association is **positive and significant but weak** (see coefficient and R^2). Refinements to try: a **time lag** (COVID surge $\rightarrow$ need a few weeks later) and **state fixed effects** to control for persistent between-state differences. These are presentation/robustness tasks, not new questions.
- The outcome is not fixed yet. `OUTCOME_INDICATOR` swaps to any single Household Pulse indicator to repeat the whole analysis; we may also broaden “need” to the **overall movement across all four indicators** or each indicator’s **average percentage change** between waves, and test which framing tracks COVID most clearly.